

1 The chain of elimination

Each of the following results removed a class of explanation and narrowed what remained. I present them in that order because the argument for the present method is cumulative.

F1. Pixel-space targets are not learnable from fMRI. My first architecture ran conditional flow matching [4, 5] over Stable-Diffusion VAE latents. It plateaued at PixCorr 0.12–0.15. A variant that derived the flow source from the conditioning collapsed outright: the training loss fell $0.70 \rightarrow 0.095$ in five epochs while reconstruction quality *fell*, because the model drove the source onto the target and made the velocity identically zero. *Consequence:* move the learning problem into a semantic space, and treat source degeneracy as a first-class hazard, a lemma I return to in F8.

F2. The token sequence admits neither a flat nor a spherical model. bigG tokens are not unit-normalised, and the decoder consumes raw values. Modelling X as a flat Gaussian ignores that CLIP semantics are carried by *direction* (every downstream metric is a cosine), and it lets the flow leave the thin shell where real tokens live. Normalising the sequence destroys the magnitudes the decoder requires, irrecoverably. *Consequence:* neither standard geometry is admissible, so I introduced the polar reparametrisation of §2, which is the contribution of this work.

F3. The manifold assumption holds only partially, and I measured it rather than assumed it. The polar split is worthwhile only if the radii are close to uninformative. On real tokens I measured a relative radius spread of 8.2% (an isotropic Gaussian would give $\approx 1.7\%$), of which only $\approx 7\%$ is positional, leaving $\approx 8\%$ genuinely content-dependent; and a pairwise direction overlap of $|\cos| = 0.35$ uncentred, falling to 0.17 after centring. *Consequence:* the radius side-model must be a real model, so I made it a per-token cross-attention head; and centring is load-bearing, which is why §4 treats it as an open measurement rather than a settled choice.

F4. The curved formulation earns its keep on stability, not only on accuracy. At matched epochs the spherical model leads the flat Euclidean baseline on the semantic axis, and the flat baseline diverged outright (NaN collapse at epoch 60; its best checkpoint scored 0.082/0.205/0.712). *Consequence:* the geometry is retained. Notably, it also dissolves two correctness hazards of the flat formulation: guidance anisotropy under standardisation, and the temptation to normalise the sequence.

F5. The decoder is not the bottleneck. Decoding *ground-truth* tokens reaches PixCorr 0.638 on the full 982 images. My prior’s anchor sits at a mean token cosine of ≈ 0.37 . *Consequence:* all remaining headroom is in the prior. This is the single measurement that determined everything after it.

F6. The low-level pathway fails from both directions, which indicts the target. MindEye2 obtains its pixel scores by fusing a semantic pathway with a low-level one that supplies the sampler’s initialisation. I implemented the same mechanism four times: mean-pooled tokens; a spatial token grid; a head reading raw voxels; and the same with L_1 on a lower-resolution target. The first three plateaued at exactly the score of predicting the per-image mean colour, and a 17.4M head scored identically to a 1.5M one (0.0699 vs. 0.0702), so capacity was never the constraint. The fourth removed the mean-seeking behaviour and overfitted instead, validation error rising monotonically $0.083 \rightarrow 0.104$. *Consequence:* underfitting and overfitting failing on the same objective indicts the *target*, not the model. A 49k-dimensional RGB image is not a learnable intermediate from 27k trials. This result also supplies the first half of my argument in §3.

F7. Capacity is the wrong lever. Widening the encoder $2048 \rightarrow 4096$ produced 0.091/0.233/0.748 and overfitted from the *first* evaluation: validation token cosine fell monotonically $0.3485 \rightarrow 0.3053$

across 150 epochs. *Consequence:* the model is data-starved, not capacity-starved. Any remaining gain must come from better conditioning of the existing problem.

F8. The generative model was being started from a strictly uninformative point. This is the observation the present work turns on. The flow transported a uniform sample $z_0 \sim \text{Unif}(\mathbb{S}^{d-1})$ onto the target. In $d = 1664$ dimensions, $\langle z_0, z_1 \rangle \sim \mathcal{N}(0, 1/d)$, so the source is $\approx 90^\circ$ from *every* target and equidistant from all of them: it carries no information whatsoever. Meanwhile my regression anchor $\mu(c)$ already stood at $\cos \approx 0.37$. *Consequence:* the transport was being asked to begin strictly behind a quantity I already possessed. Transporting from the anchor instead (§2.3) makes $v \equiv 0$ the identity map, so the prior is *bounded below* by the anchor and can only refine it.

F9. The training objective was mis-allocating capacity across time. Having fixed the source, I re-derived the objective and found the velocity parametrisation ill-conditioned: it implicitly weights endpoint error by $1/(1-t)^2$, concentrating capacity where the problem is trivial and starving $t \rightarrow 0$, which is where inference begins (§2.4). *Consequence:* predict the endpoint, not the velocity.

F10. No discriminative signal reached the decoder’s input. The contrastive term acted on a pooled auxiliary projection that the decoder never sees. Every one of the 425,984 numbers the decoder consumes was supervised by mean-seeking regression alone, the same condition that produced the collapses in F6. *Consequence:* apply the contrastive loss to the flattened predicted token grid (§2.5).

F1–F7 establish that the prior is the bottleneck and that neither capacity nor pixel-space supervision will move it. F8–F10 are the response, and together they define the method below.

2 The resulting method

2.1 Polar reparametrisation onto the oblique manifold

I split each token about a fixed mean,

$$X_i = \mu + r_i z_i, \quad r_i = \|X_i - \mu\| \in \mathbb{R}_+, \quad z_i = \frac{X_i - \mu}{r_i} \in \mathbb{S}^{d-1}, \quad d = 1664. \quad (1)$$

The map is a bijection, so the decoder still receives exact raw tokens. The directions live on the *oblique manifold* $\mathcal{M} = (\mathbb{S}^{d-1})^N$, $N = 256$, and are modelled by Riemannian flow matching [3]; the radii are handled by a light side-model. I state plainly that this is an *imposed* reparametrisation rather than native geometry, and F3 is its empirical justification.

I further factor the sequence against the pooled embedding $c_{\text{cls}} \in \mathbb{S}^{1279}$, a near-complete semantic summary: $p(X | c) = p(c_{\text{cls}} | c) p(X | c, c_{\text{cls}})$, sampling the global direction first so that it steers local detail.

2.2 Geodesic transport and its singularities

For endpoints separated by $\theta = \arccos \langle z_0, z_1 \rangle$ the conditional path is the great-circle interpolant, with target velocity

$$u_t = \frac{\theta}{\sin \theta} \left(-\cos((1-t)\theta) z_0 + \cos(t\theta) z_1 \right) \in T_{z_t} \mathbb{S}^{d-1}, \quad \|u_t\| = \theta. \quad (2)$$

Outputs are tangent-projected and integration uses the exponential map, so iterates stay on $\mathcal{M}$ exactly. Three obstacles required care:

- **Removable singularities at both ends.** The $1/\sin \theta$ factor is singular for coincident *and* antipodal endpoints. Both limits exist; a naive implementation returns 0/0 and propagates NaN through the batch. I guard each expression on its degenerate set and substitute the analytic limit.
- **Conditioning of the logarithmic map.** Writing $\theta = \arccos \langle z, y \rangle$ needs the argument clamped away from ± 1 , and that clamp puts a *floor* under $\text{Log}_z(y)$: at $y = z$ it returns a small vector in an arbitrary direction. This is harmless when Log merely measures a displacement, but fatal in §2.4, where it is evaluated at coincidence and integrated. The stable form, $\theta = \text{atan2}(\|\mathcal{P}_z y\|, \langle z, y \rangle)$, is exact at $\theta = 0$, well-conditioned near π , and needs no clamp.
- **Out-of-distribution query times.** Training samples t from a logit-normal law with almost no mass at the endpoints, yet a naive integrator evaluates the field at exactly $t=0, 1$. On a manifold this is worse than in flat space: a bad velocity can push the iterate where the retraction is ill-conditioned. I mix uniform t into training and clamp query times.

2.3 An informative source, and how to calibrate it

Following F8, the source becomes a *wrapped Gaussian* posterior centred on the regression anchor:

$$q(z_0 \mid y) = \text{Exp}_\mu(\mathcal{P}_\mu \varepsilon), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d). \quad (3)$$

The derivation permits this; only the instantiation fixed the source. The factorisation of the posterior constrains the law at $t=1$. The flow-matching construction that realises it does not constrain the law at $t=0$.

Proposition 1 (Admissibility of an informative source). *Let $p_0(\cdot \mid y)$, $p_1(\cdot \mid y)$ be laws on $\mathcal{M}$ and let the coupling be $\pi_y = p_0(\cdot \mid y) \otimes p_1(\cdot \mid y)$. Write γ_t for the geodesic interpolant, $p_t = \text{law}(\gamma_t)$ and $u_t(z \mid y) = \mathbb{E}_{\pi_y}[\dot{\gamma}_t \mid \gamma_t = z]$. Then for all θ ,*

$$\nabla_\theta \mathbb{E}_{t, \pi_y} \|v_\theta(\gamma_t, t) - \dot{\gamma}_t\|^2 = \nabla_\theta \mathbb{E}_{t, p_t} \|v_\theta(z, t) - u_t(z \mid y)\|^2. \quad (4)$$

Proof. Expand both squares. The terms $\mathbb{E}\|v_\theta\|^2$ agree because $\gamma_t \sim p_t$ by construction; $\mathbb{E}\|\dot{\gamma}_t\|^2$ and $\mathbb{E}\|u_t\|^2$ are θ -independent. For the cross terms, condition on γ_t and apply the tower property: $\mathbb{E}\langle v_\theta(\gamma_t), \dot{\gamma}_t \rangle = \mathbb{E}[\langle v_\theta(z), u_t(z \mid y) \rangle]$. The objectives differ by a θ -independent constant. $\square$

Remark. The proof uses only that γ_t has law p_t and that u_t is a conditional expectation. It never uses $p_0 = \text{Unif}(\mathbb{S}^{d-1})$, and it never uses $z_0 \perp y$. Since $z_0 = \text{Exp}_\mu(\sigma \varepsilon)$ with ε drawn independently, we have $z_0 \perp z_1 \mid y$: conditionally on y the coupling remains an independent product, so none of the machinery required for genuinely dependent couplings is invoked. The $t=1$ marginal, the only quantity the posterior factorisation constrains, is unchanged.

Proposition 1 says the change is *permitted*. The next says a particular version of it is *obligatory*.

Proposition 2 (The source must not be deterministic). *Let $v_\theta(\cdot, t)$ be locally Lipschitz, so the flow map $\psi_{0 \rightarrow 1} : \mathcal{M} \rightarrow \mathcal{M}$ is a diffeomorphism. If $p_0(\cdot \mid y) = \delta_{\mu(y)}$ then $p_1^\theta(\cdot \mid y) = \delta_{\psi(\mu(y))}$ for every θ ; consequently $\text{KL}(p_1 \parallel p_1^\theta) = \infty$ whenever $p_1(\cdot \mid y)$ is non-atomic, and no choice of θ improves it.*

Proof. A diffeomorphism's pushforward preserves absolute continuity and singularity, and $\psi_\# \delta_x = \delta_{\psi(x)}$. A point mass is mutually singular with every non-atomic measure. $\square$

Remark. Conversely a wrapped Gaussian with $\sigma > 0$ has a smooth strictly positive density on the compact manifold $\mathcal{M}$; by Moser's theorem any two such densities are related by a diffeomorphism, so the hypothesis class is adequate.

This is the reason the width must be trainable rather than nominal, and it sharpens the framing of the whole project: reconstruction was posed probabilistically precisely because the inverse map does not exist. A deterministic anchor would collapse the posterior to a point mass and void that premise, leaving only the decoder’s own sampler, which sits downstream of the latent and structurally cannot express uncertainty *about* it. The variational source is therefore the faithful arm and the deterministic one is the ablation, which is the reverse of how the two appear at first glance.

How much the change is worth, before any experiment. By Proposition 1 the attainable loss floor is the Bayes risk $L^*(t) = \mathbb{E}_{p_t}[\text{Var}(\dot{\gamma}_t \mid \gamma_t)]$; write $R(t) = L^*(t)/\text{Var}(\dot{\gamma}_t)$ for the fraction of the regression target that *no* network can predict. In the tangent-Gaussian model, with $a = 1 - t$, $b = t$ and $\rho = \sigma/\tau$ the ratio of source width to anchor error,

$$R(t) = 1 - \frac{(b - a\rho^2)^2}{(a^2\rho^2 + b^2)(1 + \rho^2)}. \quad (5)$$

A uniform draw sits at geodesic distance $\pi/2$ from any fixed μ , so the previous source is the member of this same one-parameter family with $\rho = (\pi/2)/\theta$; the present one has $\rho = 0.15/\theta$.

t	0.10	0.25	0.50	0.75	0.90	$\mathbb{E}_t[R]$
uniform source ($\rho = 1.32$)	0.448	0.610	0.928	0.946	0.767	0.755
anchored source ($\rho = 0.13$)	0.683	0.218	0.061	0.028	0.019	0.194

Monte-Carlo on the true geometry ($d = 128$, $N = 4 \times 10^4$, linear predictor, which upper-bounds R , so the gap is a lower bound) gives 0.988 against 0.098 at $t = 0.5$. Under the previous source roughly 99% of the regression target was unpredictable from (z_t, t) : the network could only learn a heavily averaged field, which is precisely the observed outcome of a 400M-parameter transport losing to a two-layer regression head. I note honestly that at small t the anchored source is *worse*, because a concentrated source reveals little about the residual early, and that this is an argument for revisiting the time sampler rather than an argument against the source.

The width is the only dial, and it trades the two propositions against each other. $\mathbb{E}_t[R] \rightarrow 0$ as $\sigma \rightarrow 0$ (perfectly learnable) exactly where Proposition 2 bites (posterior collapses); $\mathbb{E}_t[R]$ increases with σ toward the uniform-source regime. The Kullback–Leibler floor sets the operating point, which makes its weight a principled quantity rather than a tuning knob:

displacement ϕ (rad)	0	0.05	0.15	0.30	0.60	1.19	$\pi/2$
$\mathbb{E}_t[R]$	0	0.065	0.194	0.371	0.630	0.784	0.755
	collapse		<i>used</i>			$\phi = \theta$	previous

Every point of the path then carries brain information, and $v \equiv 0$ is the identity flow. Two subtleties, both high-dimensional in origin:

- **The width must not match the anchor error.** A draw is displaced in a *random* tangent direction while the target sits in one specific direction; for large d these are near-orthogonal and $\cos(z_0, z_1) \approx \cos \phi \cos \theta$, with ϕ the displacement. At the measured operating point ($\cos \theta = 0.37$, $\theta = 68.3^\circ$), setting $\phi = \theta$ would drive $\cos(z_0, z_1)$ to 0.126, worse than the anchor itself. Setting $\phi = 0.15$ rad costs $0.370 \rightarrow 0.366$, i.e. nothing measurable, while still giving the field a neighbourhood on which to be defined.
- **Angles, not coordinate variances.** For an isotropic tangent Gaussian $\mathbb{E}\|\mathcal{P}_\mu \varepsilon\| \approx \sigma\sqrt{d-1}$, and on the sphere that norm *is* the distance travelled; hence $\sigma = \phi/\sqrt{d-1}$. A width specified as a raw σ silently changes meaning with the token dimension, a factor of $\sqrt{1663} \approx 41$ here. Every angular hyper-parameter is therefore expressed in radians and converted internally.

What may carry gradient. F1’s lemma applies directly. If the flow loss pushes on the base point μ , the model shortens its own transport by degrading the anchor. If nothing in the source carries gradient, the width is fixed by its prior and the variational structure is decorative. The correct asymmetry is: *base point detached* (it is supervised by its own cosine loss), *width trainable*. The flow then pulls σ down and a Kullback–Leibler term floors it; per tangent dimension,

$$\text{KL} = \frac{1}{2} \left[\frac{\sigma^2}{\sigma_p^2} - 1 + 2 \log \frac{\sigma_p}{\sigma} \right], \quad \frac{\partial \text{KL}}{\partial \log \sigma} = \frac{\sigma^2}{\sigma_p^2} - 1, \quad (6)$$

whose unique stationary point is $\sigma = \sigma_p$. The equilibrium between that floor and the flow’s downward pressure makes the posterior a genuine model of anchor uncertainty.

Guidance becomes ill-posed. Classifier-free guidance contrasts a conditional velocity against an unconditional one. When z_0 is itself brain-derived, the “unconditional” branch still starts from a brain-dependent point, so the branches do not differ in the way guidance assumes. I therefore set the guidance scale to 1 under an informative source.

Staging. The preceding analysis assumes a *stationary* pair (p_0, p_1) . Detaching the base point removes the degenerate gradient of F1, but μ still moves under its own cosine loss, so the source drifts beneath the transport. I therefore train the encoder and anchor to convergence, freeze them, then train the transport alone against a fixed source. Beyond making Proposition 1 apply as stated, this is a substantial economy: every subsequent ablation of the transport reuses one frozen first stage, and the arms become exactly comparable because the source is bit-identical across them. Joint training is retained as an ablation.

2.4 Endpoint rather than velocity parametrisation

This follows from an identity, not an experiment. For the geodesic path,

$$u_t = \frac{\text{Log}_{z_t}(z_1)}{1 - t}, \quad (7)$$

verified numerically at $d = 1664$ to 4×10^{-8} – 2×10^{-6} , i.e. float32 round-off. The two parametrisations describe the *same* field; only the loss geometry differs. Substituting into the velocity mean-squared error,

$$\|v_\theta - u_t\|^2 = \frac{\|\text{Log}_{z_t}(\hat{z}_1) - \text{Log}_{z_t}(z_1)\|^2}{(1 - t)^2}, \quad (8)$$

so training on velocity weights *endpoint* error by $1/(1 - t)^2$. Two consequences, both measured:

t	0.1	0.5	0.9	0.98
implicit endpoint weight $1/(1 - t)^2$	$1.2\times$	$4\times$	$100\times$	$2500\times$
anchor error θ	90.0°	68.3°	30.0°	15.0°
(cosine loss) / (velocity loss)	$674\times$	$738\times$	$813\times$	$827\times$

The first row is backwards: near $t=1$ recovering z_1 from z_t is nearly trivial, while near $t=0$ the inversion is ill-conditioned and the model must genuinely use the brain signal, and $t=0$ is where inference starts. The second follows from $\|u_t\| = \theta$ in (2): since the mean-squared error averages over d coordinates, the flow term sits three orders of magnitude below its neighbours, and the imbalance *tightens* as the model improves.

Resolution. The network emits a residual toward the endpoint,

$$\hat{z}_1 = \frac{z_t + f_\theta(z_t, t, c)}{\|z_t + f_\theta(z_t, t, c)\|}, \quad v_\theta = \frac{\text{Log}_{z_t}(\hat{z}_1)}{1 - t}, \quad (9)$$

trained on $1 - \cos(\hat{z}_1, z_1)$, uniform in t . The $1/(1 - t)^2$ re-weighting disappears; the flow term acquires the same natural scale as the anchor loss (measured ratio ≈ 7 rather than $\approx 1/740$), making their relative weight an interpretable hyper-parameter; and because the output layer is zero-initialised, $\hat{z}_1 = z_t$ at initialisation, so $v \equiv 0$ and the identity-flow bound of F8 holds *by construction*.

2.5 Discriminative supervision on the decoder path

Following F10, the contrastive loss is additionally applied to the flattened 256×1664 predicted grid rather than to a pooled projection. Pooling would discard precisely the per-position structure the term exists to supervise. Contrastive objectives are the natural counterweight to mean-seeking regression because they are *relative*: a prediction resembling the dataset average scores at chance against in-batch distractors, however good its cosine to its own target.

2.6 Exposure bias in the two-head factorisation

The patch head sees the true c_{cls} during training and a sampled estimate at inference. Since c_{cls} nearly determines the image, a head given it for free learns to route around the brain tokens. I treat this as scheduled sampling and support teacher forcing, a geodesic rotation of the truth by a calibrated angle, and true sampled conditioning. The rotation angle is calibrated directly from the logged agreement between the sampled and true global direction, which is another place where expressing perturbations as angles rather than coordinate variances is necessary rather than merely tidy.

3 Why PixCorr is not a sound optimisation target

I report PixCorr because the literature does, but I do not let it drive design. Three independent arguments agree.

1. Theory: distortion versus perception. PixCorr is a squared-error criterion, uniquely maximised by the posterior mean $\mathbb{E}[x | y]$. Under fMRI conditioning that posterior is broad, since the measurement does not determine high-frequency content, so $\mathbb{E}[x | y]$ is a *blur*: not a plausible image, yet the PixCorr optimum. Conversely a model sampling correctly from $p(x | y)$ incurs roughly twice the posterior mean’s squared error. This is the perception-distortion trade-off [2]: **a perfect generative model is penalised relative to a blur**. Optimising PixCorr selects against the behaviour I want, and it is exactly the failure observed four times in F6.

2. Measurement: PixCorr scores the initialisation, not the decoding. Holding the semantic content fixed at ground truth and varying only how much low-frequency information the sampler is handed:

Decoder input (semantics identical throughout)	PixCorr	SSIM	CLIP
GT tokens, noise initialisation (full 982)	0.638	0.249	n/a
GT tokens, GT-blur init, strength 1.0	0.730	0.353	≈ 1.0
GT tokens, GT-blur init, strength 0.7	0.870	0.444	≈ 1.0
GT tokens, GT-blur init, strength 0.5	0.913	0.478	≈ 1.0

PixCorr moves $0.638 \rightarrow 0.913$ on initialisation alone. It is therefore largely a measure of the sampler’s starting point, not of what was decoded from the brain.

3. Operation: the trade-off is a one-parameter frontier with closed-form ends. One checkpoint, full 982 images, two decode configurations:

Decode configuration	PixCorr	SSIM	CLIP-2way
regression point estimate, guidance 1.0	0.164	0.348	0.703
blended prior sample, guidance 1.0	0.131	0.276	0.738

The same weights give either the better pixel score or the better semantic score, with no retraining; the low-variance point estimate, which is the least generative option, wins on PixCorr, exactly as argument 1 predicts.

Because the transport now begins at the anchor, this dial ceases to be an ad-hoc blend and becomes a single integer. Averaging K independent source draws on the sphere, $\hat{z}_K = \|K^{-1} \sum_k \psi_{0 \rightarrow 1}(z_0^{(k)})\|^{-1} K^{-1} \sum_k \psi_{0 \rightarrow 1}(z_0^{(k)})$, interpolates between a calibrated posterior sample and the posterior mean, with both endpoints fixed in closed form by Proposition 3.

Proposition 3 (Endpoints of the multi-sample frontier). *Fix y and suppose the transport is calibrated, i.e. $\psi_{0 \rightarrow 1}(z_0) \sim p_1(\cdot | y)$ with $z_1 \sim p_1(\cdot | y)$ drawn independently. Let $m = \mathbb{E}[\psi_{0 \rightarrow 1}(z_0) | y]$ and assume $m \neq 0$. Then*

$$\mathbb{E}[\cos(\hat{z}_1, z_1)] = \|m\|^2, \quad \lim_{K \rightarrow \infty} \mathbb{E}[\cos(\hat{z}_K, z_1)] = \|m\|. \quad (10)$$

Proof. For $K = 1$, independence gives $\mathbb{E}\langle \psi(z_0), z_1 \rangle = \langle \mathbb{E}\psi(z_0), \mathbb{E}z_1 \rangle = \|m\|^2$. For $K \rightarrow \infty$ the strong law gives $K^{-1} \sum_k \psi(z_0^{(k)}) \rightarrow m$ almost surely, hence $\hat{z}_K \rightarrow m/\|m\|$ by continuity of $u \mapsto u/\|u\|$ at $m \neq 0$; then $\mathbb{E}\langle m/\|m\|, z_1 \rangle = \langle m, m \rangle / \|m\| = \|m\|$. $\square$

Since $\|m\| \leq 1$, the frontier is monotone: $K = 1$ is maximally generative and should favour the ranking statistics, while $K \rightarrow \infty$ recovers the posterior mean and should favour PixCorr. Substituting the measured anchor $\|m\| = 0.370$ gives the predicted frontier

K	1	2	4	8	16	32	∞
predicted $\mathbb{E}[\cos(\hat{z}_K, z_1)]$	0.137	0.181	0.231	0.277	0.314	0.338	0.370
predicted dispersion rel. to data	1.000	0.938	0.841	0.714	0.572	0.436	0

These are predictions of the theory, not measurements; the frontier has not yet been traced. Its value is that it requires no fitting, since both ends follow from $\|m\|$ alone, so it converts the perception-distortion trade-off from an argument about metric validity into a falsifiable curve traced by one trained model, and it replaces the blend weight I was previously sweeping.

What I use instead.

- **Ranking statistics** (two-way identification, k -way retrieval). Writing a predictor as $\hat{x} = ax + b\bar{m} + e$ (signal, shared component, idiosyncratic error), a pointwise criterion is increasing in the shared component b whereas a centred pairwise criterion is invariant to it. Ranking statistics cannot be gamed by predicting the dataset average, which is the specific failure mode of F6.
- **Angular fidelity added over the anchor.** Because the transport now begins at the anchor, the gain of the flow over its own starting point is directly measurable, and isolates the generative component from the regression baseline. It needs no diffusion pass, so I log it on both splits at every evaluation; the train/validation gap then separates underfitting from overfitting per component.
- **Selection on a ranking statistic.** I previously selected checkpoints on a pointwise token cosine and observed it move *opposite* to decoded semantic quality over a hundred epochs of one run. Selection now uses the two-way statistic on the quantity the decoder receives.

This is not an argument that low-level fidelity is unimportant. It is an argument that PixCorr is the wrong instrument for it, and that the correct route is structural, namely a working low-level pathway supplying the sampler’s initialisation, rather than tuning a semantic model against a criterion it cannot satisfy without becoming a blur.

4 Standing and the immediate question

Method	PixCorr $\uparrow$	SSIM $\uparrow$	CLIP $\uparrow$	Incep $\uparrow$
Ours, current cluster run			0.942	0.956
Ours, regression point estimate	0.164	0.348	0.703	
Ours, blended prior sample	0.131	0.276	0.738	
Takagi & Nishimoto 2023 [8]	0.246	0.410	0.821	
Brain-Diffuser [6]	0.273	0.365	0.909	0.913
MindEye2 [7]	0.322	0.431	0.930	0.954

On the semantic axis the current cluster run reaches Inception 95.6% against MindEye2’s 95.4%, and CLIP 94.2% against 93.0%. The Inception margin is within run-to-run noise and I do not claim it. The CLIP margin is the meaningful one: 1.2 points over MindEye2 and 3.3 over Brain-Diffuser is a real gap on that metric, and it is exactly the axis this document argues a brain-conditioned prior should be judged on. It is also the axis on which F4 already showed the curved formulation leading the flat baseline, so the result is consistent with the geometry being the operative cause rather than an incidental one.

The earlier rows remain in the table because they are the two decode configurations that trace the trade-off of §3, and because the pixel-level gap is unchanged by the semantic result: I still have no working low-level pathway, exactly as F6 predicts, while MindEye2’s PixCorr and SSIM come from fusing one with its semantic pathway.

Established. The curved formulation trains stably where the flat baseline diverges and leads it on the semantic axis (F4). The decoder is not the bottleneck (F5). Capacity is not the lever (F7).

The one open question. Whether the transport, now started from an informative source (F8) and trained under a well-conditioned objective (F9), improves on its own anchor. This is a single scalar, it is now directly observable, and the current experiment is designed around nothing else.

Two caveats I am not yet in a position to close. First, the centring vector is a single element of $\mathbb{R}^{1664}$ averaged over images *and* over all 256 positions; if bigG tokens carry a position-specific common mode, each centred direction retains a component that is identical across images and predictable without any brain data, yet counted in full by every cosine I report. Per-position centring and the diagnostic that sizes this effect are implemented but not yet run, so my anchor cosines should be read as upper bounds. Second, the pixel-level rows are a four-subject joint model using per-subject input projections over a shared trunk, while my earlier single-subject baseline was substantially stronger; a single-subject control is required before attributing the difference to the prior.

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